MIL-L-48772A (AR)
12 May 1981
SUPERSEDES
MIL-L-48772 (PA)
19 June 1975

MILITARY SPECIFICATION

LITHIUM
(For Use in Ammunition)

This specification approved for use by the US Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions and packaging for the manufacture of two grades of lithium for use in Vanadium Pentoxide (V_2O_5) and Thionyl Chloride ($SOCl_2$), reserve type batteries (see 6.4).

1.2 Classification. Lithium shall be of the following grades specified: (see 6.1).

Grade A - Lithium with Sodium Content 0.015 percent, maximum
Grade B - Lithium with Sodium Content 0.05 percent, maximum

2. APPLICABLE DOCUMENTS

2.1 Issue of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specfied herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

FSC: 6810

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection By Attributes
MIL-STD-129 - Marking for Shipping and Storage
MIL-STD-1218 - ACS Chemicals

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement function should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ASTM-E300 Sampling Industrial Chemicals

ASTM-D2905 Statements on Number of Specimens Required to Determine the Average Quality of a Textile Material

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19013).

(Technical society and technical associations specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using federal agencies).

3. REQUIREMENTS

3.1 Material. - The material shall be lithium conforming to the following requirements: (see 6.3 and 6.4).

3.1.1 Form. The material shall have a silver metalic luster. No oils or other protective coatings shall be on the metal surface. The material shall be examined as specified in 4.6.1.

3.1.2 Chemical analysis. The material shall conform with the following table when analyzed in accordance with the applicable paragraphs:

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TABLE 1
Requirements for Lithium

Chemical Property	Requirement		Test Paragraph
Grade	A	B	
Lithium, %, min	99.9	99.8	4.6.4
Sodium, %, max	0.015	0.05	4.6.2
Potassium, % max	0.01	0.01	4.6.2
Calcium, %, max	0.02	0.02	4.6.2
Chlorides plus Iron % max	0.06	0.06	4.6.3

3.2 First article inspection. This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 6.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First Article Inspection (see 4.3)
b. Quality Conformance Inspection (see 4.4)

4.3 First article inspection

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article shall consist of approximately 5 grams of lithium obtained by sampling as described in 4.5.1 (smaller weight samples may be taken that are consistent with the requirements of the test). The sample shall be obtained from a production lot which has been produced by the contractor using the same production processes, procedures and equipment as will be used in fulfilling the contract. All materials, shall be obtained from the same sources of supply as will be used in regular production (see 6.3).

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4.3.2 Inspections to be performed. The sample will be subjected by the Government to any or all of the examinations for tests specified in 4.6 of this specification.

4.3.3 Rejection. See MIL-A-48078.

4.4 Quality Conformance Inspection.

4.4.1 Lot Formation. A lot shall consist of one or more batches of lithium produced by one manufacturer, in accordance with the same specification, or same specification revision, under one continuous set of operating conditions. Each lot shall consist of that quantity of lithium that has been subjected to the same unit chemical or physical process intended to make the final product homogeneous (See MIL-A-48078). The product shall be submitted for inspection in accordance with MIL-STD-105.

4.5 Testing (see 6.2) PRECAUTION. This specification covers sampling and testing of toxic or hazardous materials. Accordingly, it is emphasized that all applicable safety rules, regulations, and procedures must be followed in handling and processing these materials.

4.5.1 Sampling. Independent and representative samples (approx 5g) shall be withdrawn from each batch or lot of lithium using the procedure specified in ASTM E300, section 25. The samples shall be tested independently in accordance with the applicable methods given in 4.6. If either sample fails to meet any of the test requirements, the batch or lot shall be rejected. The samples shall be subjected to the tests listed below:

Table II

Inspection/Requirement	Defect Classification
Form (see 3.1.1)	Minor
Chemical Analysis (see 3.1.2)	Major

4.5.2 Inspection equipment. The contractor shall use commercial inspection equipment for all tests and examinations specified in para. 4.6. He is also charged with the responsibility of assuring that proper calibration procedures are followed. Government approval of all inspection equipment is required prior to its use.

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4.6 Test methods and procedures. (see 6.2 and 6.5) The tests to determine compliance with the requirements given in Table I shall be performed on the samples obtained in 4.5.1. All tests given in this section shall be performed using prescribed procedures for replicate determinations given in standard analytical chemistry textbooks or ASTM D2905. Also, unless otherwise specified herein, all chemicals shall be reagent grade or ACS Grade in accordance with MIL-STD-1218.

4.6.1 Form. After performing the analyses given in 4.6.2 take the necessary sample and spread out in a thin layer on a sheet of white paper. Visually inspect the sample to determine compliance with 3.1.1.

4.6.2 Calcium, Potassium and Sodium

4.6.2.1 Standards. Standard calcium solution (50 ppm). Weigh 0.6243 g of Calcium Carbonate ($CaCO_3$ into a 500 ml volumetric flask, add water and just enough Conc. Hydrochoride Acid HCl to dissolve the $CaCO_3$. Dilute to mark with water and mix. Transfer 50 ml aliquot to a 500 ml volumetric flask and dilute to mark.

Standard potassium solution (50ppm). Weigh 0.4765 g of Potassium Chloride (KCl) which has been previously dried at 110°C into a 500 ml volumetric flask, dissolve in water and make to mark and mix. Transfer 50 ml aliquot to a 500 ml volumetric flask and dilute to mark.

Standard sodium solution (50 ppm). Weigh 0.6355 g of NaCl which has been previously dried at 110°C into a 500 ml volumetric flask, dissolve in water and made to mark and mix. Transfer 50 ml aliquot to a 500 ml volumetric flask and dilute to mark.

Working standards for potassium and sodium. Into individual 500 ml volumetric flasks, place the following:

Flask #	ml Stock KCl	ml Stock Na Cl	Conc. ml HCl
1 (o ppm)	0	0	2
2 (1 ppm)	10	10	2
3 (2 ppm)	20	20	2
4 (3 ppm)	30	30	2
5 (4 ppm)	40	40	2

Dilute all to the mark and store in polyethylene bottles. The standards will contain K and Na from 0 to 4 ppm.

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Working standards for calcium. Into individual 500 ml volumetric flasks, place the following:

Flask #	ml of stock Ca CO_3	g KCl	ml conc HCl
1 (0 ppm)	0	0.1	2
2 (1 ppm)	10	0.1	2
3 (2 ppm)	20	0.1	2
4 (3 ppm)	30	0.1	2
5 (4 ppm)	40	0.1	2

Dilute all to the mark and store in polyethylene bottles. The standard will contain Ca from 0 to 4 ppm.

4.6.2.2 Procedure Prepare cleaned 400 ml plastic beakers by washing with HCl and rinsing with demineralized water. Weigh to 0.01 g about 2 grams of sample into the beakers, cover with a watch glass and add about 100 ml of demineralized water. After the sample is in solution, neutralize with HCl using phenolphthalein as indicator plus 4 ml in excess of the HCl. Quantitatively transfer the solution to clean 1000 ml volumetric flasks and dilute to the mark with demineralized water. Burn for potassium against standards using either Emission spectroscopy or Atomic Absorption (a.A.) and calculate concentration by an approved method. Make a 1 to 1 dilution of the sample and burn for sodium against standards using either Emission spectroscopy or A.A. and calculate concentration by an approved method.

Using the dilution for sodium, add about 0.2 g/liter of solid KCl. After dissolving, mix and burn for calcium by A.A. (acetylene/nitrous oxide flame) using the standards made as above. Calculate the concentration by any approved method.

4.6.2.3 Calculations

$$\text{ppm K} = \frac{\text{ppm K in solution} \times 1000}{\text{Wt of sample}}$$

$$\text{ppm Na} = \frac{\text{ppm Na in solution} \times 2000}{\text{Wt of sample}}$$

$$\text{ppm Ca} = \frac{\text{ppm Ca in solution} \times 2000}{\text{Wt of sample}}$$

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4.6.3 Chloride plus iron.

4.6.3.1 Chlorides.

4.6.3.1.1 Standards. Standard Chloride solution (0.001 N HCl). Dilute 10 ml of 0.10 N HCl to 1000 ml. This solution is 0.001 N HCl and will contain 0.0355 mg/ml of chlorides.

Silver nitrate solution (5%) Dissolve 5.0 g of silver nitrate in 100 ml water.

Chloride working curve. Into individual clean 100 ml volumetric flasks add:

Flask	ml Cl Solution	ml HNO_3	ml Ag NO_3	mg Cl Present
1	0	4	3	0
2	2	4	3	0.071
3	4	4	3	0.142
4	6	4	3	0.213
5	8	4	3	0.283
6	10	4	3	0.355

Dilute to the mark, allow to stand five minutes in a dark place and determine turbidity (hellige, DRT 100, Spectronic 20 or other instrument). Plot turbidity units versus mg Cl.

4.6.3.1.2 Procedure. Weigh about 0.4g of the lithium metal sample and transfer into a clean 400 ml plastic beaker. Add about 50 ml demineralized water into; and when reaction ceases, neutralize with conc. HNO_3 using phenolphthalein as indicator, plus 4 ml of the HNO_3 excess. Transfer the solution into a clean 100 ml volumetric flask, add 3 ml $AgNO_3$ solution, make up to mark mix and let stand for five minutes in the dark. Run on turbidimeter against a blank made up with water and reagents. Determine mg contained Cl from previously drawn working curve.

4.6.3.1.3 Calculation for Cl

$$\text{ppm Cl} = \frac{\text{mg Cl X 1000}}{\text{wt. sample}}$$

4.6.3.2 Iron

4.6.3.2.1 Standards and Solutions. Standard iron solution (0.01 mg/ml) - Weigh 0.1000 g of pure iron metal into a one-liter volumetric flask. Add about 50 ml demineralized

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water and 25 ml of concentrated HCl. When the iron has dissolved, dilute to mark with demineralized water and mix. Transfer a 50 ml aliquot to 500 ml volumetric flask, dilute to mark and mix. This solution will contain 0.01 mg/ml iron.

1,10-phenanthroline (2.5%). Disolve 2.5 g of 1.10 - phenanthroline in 100 ml of ethanol.

Hydroxylamine hydrochloride - (10%). Dissolve 50 g of hydroxylamine hydorchloride in 500 ml of water.

Sodium acetate - (saturated). Add about 100 ml water to a reagent grade sodium acetate bottle.

4.6.3.2.2 Establishing standard working curve. Pipette 0, 1, 3, 5, 8 and 10 ml aliquots of 0.01 mg/ml iron solution into 100 ml volumetric flasks and dilute to about 50 ml. Add 5 ml of hydroxylamine hydrochloride to flask and a small square of Congo Red paper. Add sodium acetate solution, with mixing, until the paper turns red. Add 1 ml of 1, 10- phenanthroline solution, dilute to volume and mix. Read absorbances at 510 mu, setting the instrument at 0 with the 0 standard. Plot absorbance vs mg Fe.

4.6.3.2.3. Procedure Weight about 2 g of lithium metal into a dry 250 ml beaker and add 50 to 100 ml hexane. Cover with a watch glass and slowly add water until sample is in solution (do in a hood). Neutralize to phenalphthalein end point with conc HCl and add several drops in excess. Heat to drive off hexane and reduce volume to about 50 ml. Cool and transfer to a 100 ml volumetric flask. Add 5 ml of hydroxylamine hydrochloride solution and a small square of Congo Red paper. Add sodium acetate solution until paper turns red. Add 1 ml of 1, 10-orthophenanthroline solution, dilute to mark and mix. Run a blank at the same time but without sample and read absorbance, at 510 mu, setting instrument to 0 with this blank.

4.6.3.2.4 Calculation for Fe. Read mg of contained Fe from previously drawn curve.

$$\text{ppm Fe} = \frac{\text{mg Fe} \times 1000}{\text{Wt sample}}$$

4.6.3.3 Calculation for Chlorides Plus Iron. Add the values in 4.6.3.1 and 4.6.3.2 to obtain the sum.

4.6.4 Determine the purity of Lithium by subtracting the percent impurities found in 4.6.2 and 4.6.3 from 100%.

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5. PACKAGING

5.1 Packing (see 6.3)

5.1.1 Unless otherwise specified, the lithium shall be packed in accordance with manufacturer's commercial practice to assure acceptance by common carrier for safe delivery at first destination for immediate use.

5.2 Marking. Unless otherwise specified, the shipping containers shall be marked in accordance with MIL-STD-129, and shall include the lot number.

6. NOTES

6.1 Ordering data. See MIL-A-48078

a. Grade

b. Grade A is intended for use in the lithium vanadium pentoxide reserve type batteries.
Grade B is intended for use in the lithium thionyl chloride reserve type batteries.

c. First article submission.

6.2 Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted thru the Contracting Officer to: Commander, ARRADCOM, ATTN: DRDAR-QAR-R, Dover, NJ 07801. This description should include but not be limited to the procedures used, the accuracy and precision of the method, test data to demonstrate the accuracy and precision and drawings of any special equipment required.

6.3 The lithium material and samples shall be shipped in sealed metal containers that maintain material under an Argon atmosphere.

6.4 Lithium which has been found satisfactory for use in lithium batteries is manufactured by Foote Mineral Company, Exton, PA 19341 and Lithium Corporation of America, Inc., P.O. Box 8, Bessemer City, NC 28016. Lithium from other sources should be evaluated by ARRADCOM, ATTN: DRDAR-QAR-R, Dover, NJ 07801 before being approved as an equivalent.

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6.5 The tests described in this specification should be performed in an inert atmosphere glove box, or in a dry room with relative humidity not to exceed 5% at ambient conditions.

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Preparing Activity
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